

Setting Up a Linux Programming Workspace

Aim	Prepare a Linux-based development environment and use it to write, compile, and run a small C program.
Tools	Oracle VirtualBox, Kali Linux virtual machine, terminal, GCC, compiler, and a text editor.
Outcome	A working virtual Linux system with verified command-line and C compilation workflows.

Environment choices

An online IDE is convenient for quick practice, whereas a local setup provides direct access to files, compilers, and system commands. A virtual machine offers an isolated Linux system without replacing the host operating system.

Workflow at a glance

- 1 Install the virtualization application on the host computer.
- 2 Import the provided Kali Linux virtual machine image.
- 3 Boot Linux and confirm that the desktop and terminal work correctly.
- 4 Create a C source file, compile it with GCC, and execute the output file.
- 5 Practise navigation, file-management, and redirection commands.

Part A - Preparing the Virtual Machine

Installation sequence

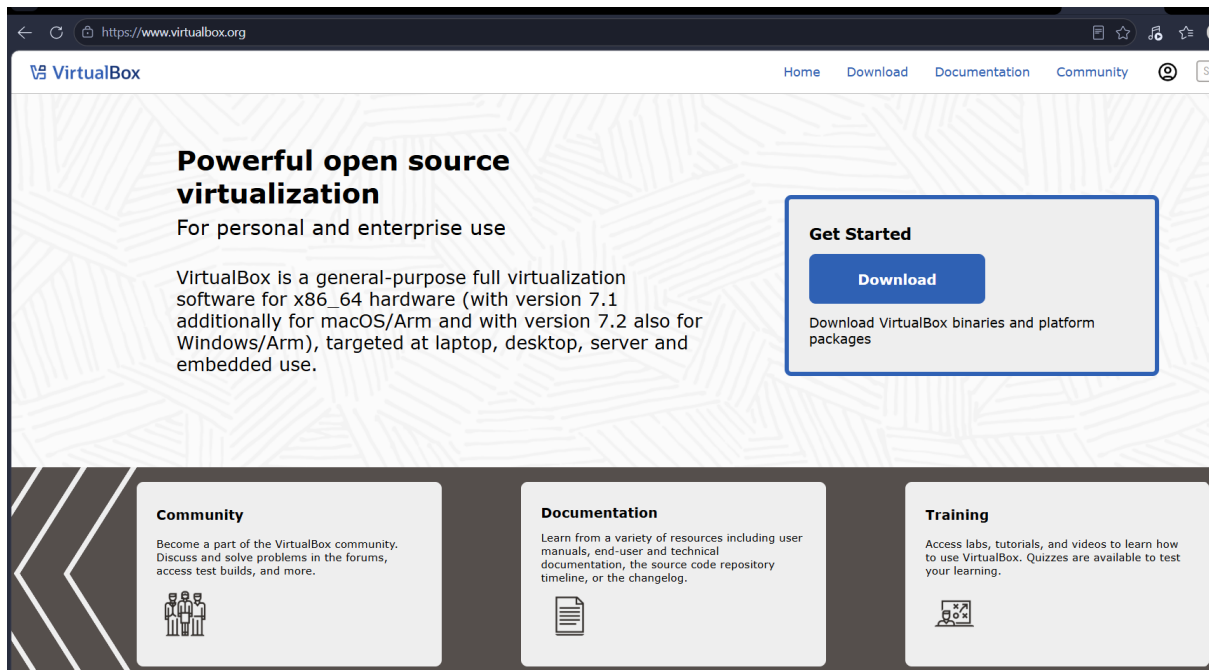
Oracle VirtualBox provides the virtualization layer. The Kali Linux appliance supplies a preconfigured guest operating system, so installation is mainly an import-and-start process.

- 1 Download VirtualBox from the official Oracle website and run the installer.
- 2 Download the prebuilt Kali Linux VirtualBox image from the official Kali website.

EXPERIMENT_1

3 Extract the archive, then use VirtualBox to add or import the virtual machine.

4 Review memory, processor, and storage settings before the first boot.

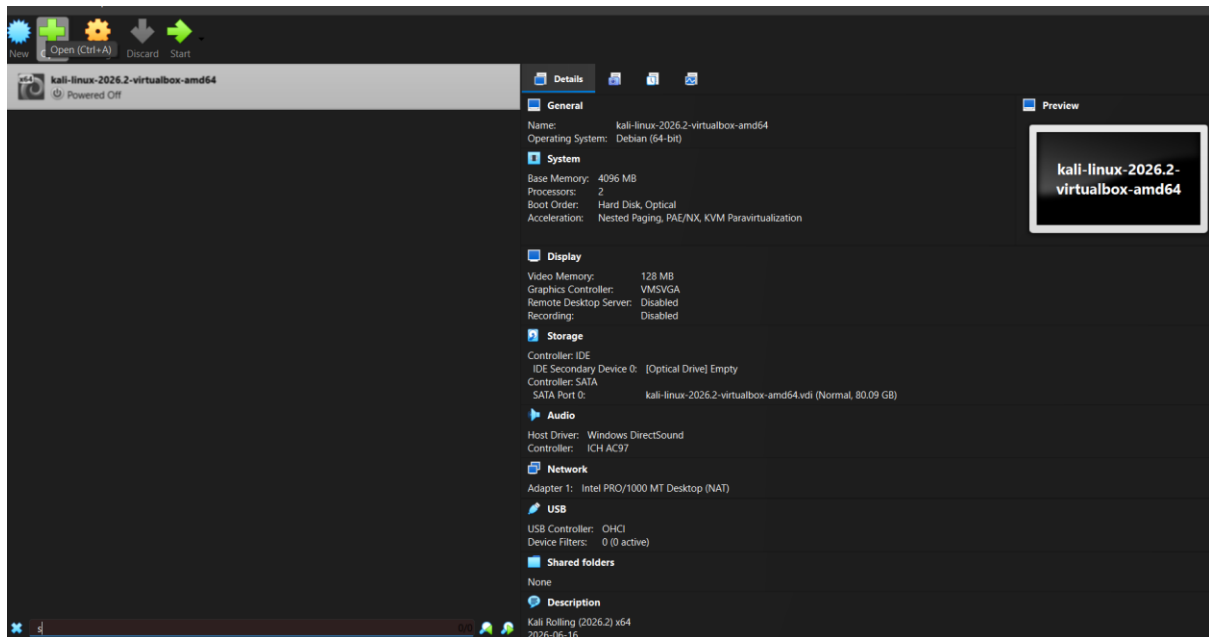


FIGURE_1 webpage to install virtual box

EXPERIMENT_1

Part A - First Boot and Verification

Starting the guest operating system After the appliance is available in VirtualBox, select it and press Start. A successful boot displays the Kali desktop. Opening a terminal confirms that the guest system is responsive and ready for programming work



1.1 click on start to boot Kali linux



1.2_enter password and start linux

Part B - Writing and Running a C Program

Source program

Program used

```
#include <stdio.h>

int main(void)
{
    printf("My Name is prabhu\n");
    return 0;
}
```

Compilation workflow

- 1 Create a file named exp1.c and enter the source code.
- 2 Compile it with gcc exp1.c -o exp1.
- 3 GCC translates the source into an executable file. 3 Run the executable using ./exp1 and verify the printed output.

Part C - Linux Command Reference

The commands below cover navigation, directory management, file operations, permissions, and output redirection. Command Practical use

Pwd- Show current directory

mkdir- Create a directory

cd -Change directory

ls- List directory contents

clear- Clear the terminal

history- Review earlier commands

man -Open a command manual

whoami- Show the active user

touch- Create an empty file

file- Identify a file type

EXPERIMENT_1

cp- Copy files or folders

mv- Move or rename an item

rm -Remove an item

more / less -Read text page by page

head / tail- View beginning or end of a file

chmod- Modify file permissions

>- Write output to a file

>> -Append output to a file

Result

A Linux programming workspace was configured successfully inside VirtualBox. The Kali Linux guest system booted correctly, a basic C program was compiled and executed using GCC, and common terminal commands were tested for navigation, file handling, permissions, and redirection. The experiment therefore demonstrates the complete beginner workflow: prepare the environment create source code - compile - execute - manage files from the terminal